

VORONEZH, Russian Federation

WMO: 341230

Lat: 51.700N

Lon: 39.217E

Elev: 489

StdP: 14.44

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.2	(c) -3.5	(d) -14.8	(e) 2.5	(f) -8.6	(g) -9.5	(h) 3.4	(i) -2.6	(j) 20.5	(k) 27.2	(l) 18.4	(m) 24.9	(n) 4.0	(o) 30	(p) 0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	89.9	68.0	86.4	66.9	83.2	65.6	70.7	84.0	69.1	81.4	67.7	79.1	6.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	99.4	74.7	65.2	94.7	73.7	63.7	89.9	72.8	34.8	84.1	33.5	81.7	32.4	79.2	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	15.4	13.6	-13.0	94.5	6.4	3.7	-17.6	97.1	-21.4	99.3	-25.0	101.3	-29.7	104.0
			-13.4	74.0	6.4	2.8	-18.0	76.1	-21.7	77.7	-25.3	79.3	-29.9	81.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.7	19.9	21.8	31.5	48.1	60.2	66.5	70.6	68.1	57.4	45.3	33.4	24.3
	DBStd	20.22	11.82	11.60	8.51	8.48	8.35	6.90	6.10	6.96	7.66	8.22	9.36	11.13
	HDD50	3956	932	790	574	136	15	0	0	0	20	192	499	798
	HDD65	7543	1397	1210	1039	509	195	64	17	45	247	610	947	1263
	CDD50	2397	0	0	1	78	331	496	638	562	243	47	1	0
	CDD65	509	0	0	0	1	45	110	191	142	20	0	0	0
	CDH74	4572	0	0	0	13	441	915	1657	1359	185	2	0	0
	CDH80	1512	0	0	0	1	105	260	599	514	33	0	0	0
	WSAvg	6.1	6.9	7.4	7.2	6.4	5.7	5.4	4.8	4.9	5.1	5.9	6.8	7.2
	PrecAvg	22.7	1.7	1.3	1.3	1.5	1.8	2.5	2.6	2.3	2.1	1.7	1.9	2.0
Precipitation	PrecMax	32.7	3.8	2.9	3.3	3.1	5.6	8.5	7.8	7.3	7.2	5.5	4.3	4.6
	PrecMin	14.9	0.3	0.0	0.2	0.1	0.3	0.2	0.1	0.1	0.2	0.0	0.1	0.5
	PrecStd	3.7	0.9	0.7	0.7	0.8	1.1	1.5	1.4	1.3	1.5	1.0	1.2	1.1
	PrecStd	3.7	0.9	0.7	0.7	0.8	1.1	1.5	1.4	1.3	1.5	1.0	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	41.4	58.0	75.9	87.2	90.4	95.5	96.3	84.2	70.3	56.1	46.5
		MCWB	38.8	38.7	45.7	55.5	65.6	65.9	69.4	67.9	61.9	57.4	51.9	44.4
	2%	DB	37.5	38.7	50.8	70.7	83.0	86.2	90.1	90.1	79.2	65.1	52.0	42.5
		MCWB	35.6	36.5	42.4	54.4	62.3	66.4	69.2	67.8	60.6	55.7	49.0	41.0
	5%	DB	35.6	36.6	46.1	66.3	79.1	82.8	86.4	85.7	74.8	60.7	48.0	39.5
		MCWB	34.0	34.6	40.0	52.4	61.3	66.0	68.3	66.4	60.0	54.0	45.3	37.8
	10%	DB	34.0	35.0	42.3	62.2	75.1	79.4	82.7	81.3	70.3	56.8	44.8	36.6
		MCWB	32.7	33.2	37.7	50.2	60.0	64.8	66.9	64.8	58.1	51.7	42.7	35.1
	0.4%	WB	39.4	39.5	47.8	58.0	67.7	71.6	73.6	71.8	65.4	60.4	52.8	45.0
		MCDB	40.5	40.7	56.1	70.3	82.5	82.6	87.7	86.9	74.3	66.6	55.4	46.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.0	37.0	43.3	55.6	65.1	69.3	71.4	69.6	62.9	57.3	49.2	41.2
		MCDB	37.0	38.3	49.2	65.8	78.2	80.5	85.3	84.2	74.0	62.9	51.9	42.5
	5%	WB	34.2	34.9	40.5	53.6	63.0	67.7	69.8	68.0	61.3	54.7	45.9	38.0
		MCDB	35.3	36.1	45.1	63.9	74.7	78.4	82.1	81.4	71.1	59.5	47.8	39.1
	10%	WB	32.7	33.5	38.2	51.5	61.2	66.0	68.3	66.4	59.6	52.2	42.9	35.2
		MCDB	33.9	34.7	41.9	60.7	72.3	76.3	79.2	78.3	68.0	56.5	44.6	36.3
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.9	11.3	16.6	18.5	17.3	17.9	19.3	16.7	11.9	8.1	8.0
		MCDBR	6.3	7.6	16.7	21.9	23.6	22.0	23.2	25.9	23.8	17.5	11.0	7.1
		MCWBR	6.2	6.8	10.4	10.7	8.8	8.3	8.1	8.9	9.9	8.9	8.9	6.6
	5% WB	MCDBR	6.3	6.9	14.7	19.4	20.6	18.7	20.1	21.1	19.5	15.3	10.0	7.0
		MCWBR	6.3	6.9	9.8	10.4	9.0	8.6	8.3	8.8	9.7	10.1	8.8	6.8
		MCWBR	6.3	6.9	9.8	10.4	9.0	8.6	8.3	8.8	9.7	10.1	8.8	6.8
Clear-Sky Solar Irradiance	taub		0.305	0.341	0.365	0.442	0.398	0.390	0.421	0.436	0.388	0.362	0.338	0.307
	taud		2.124	2.026	2.099	2.088	2.315	2.353	2.274	2.216	2.332	2.353	2.276	2.176
	Ebn at Noon		212	236	258	251	272	274	263	251	251	233	201	190
	Edh at Noon		24	35	40	46	38	37	40	40	32	26	21	20
All-Sky Solar Radiation	RadAvg		303	631	1065	1324	1815	1889	1853	1605	1061	560	246	175
	RadStd		69	85	170	118	148	166	151	135	142	78	51	49

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.49	N/A	N/A	N/A	N/A	N/A	N/A	-328	+193	N/A
(47)	Regional (6 neighbors)	+1.33	N/A	N/A	N/A	N/A	N/A	N/A	-331	+209	N/A

Nomenclature: See separate page